INTEGRATION TESTS WITH @SPRINGBOOTTEST, SPRING SECURITY, AND TESTCONTAINERS

The integration tests we wrote in the previous chapters will not work anymore for two reasons. First, all `POST`, `PUT`, and `DELETE` requests will fail because we are not providing any valid OAuth2 Access Token. Even if we were, there is no Keycloak up and running, required by Spring Security to fetch the public keys used to validate the Access Tokens.

You can verify the failure by running the following command from the Catalog Service root folder:

```
$ ./gradlew test --tests CatalogServiceApplicationTests
```

We have already seen how to use Testcontainers to write integration tests against data services like a PostgreSQL database, making our tests more reliable and ensuring environment parity. In this section we'll do the same for Keycloak.

Let's start by configuring a Keycloak container via Testcontainers. Open the `CatalogServiceApplicationTests` class and add the following setup.

Listing 12.13 Setup for a Keycloak test container

```
@SpringBootTest(webEnvironment = SpringBootTest.WebEnvironment.RANDOM_PORT)
@ActiveProfiles("integration")
@Testcontainers                                                    ❶
class CatalogServiceApplicationTests {

  @Autowired
  private WebTestClient webTestClient;
```